

TODD RUSSELL

Eagle Scout · Runner · Scuba Diver

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Authorized to work in the U.S. without sponsorship.

PROFESSIONAL STATEMENT

Mechanical engineer with full design ownership experience, from concept modeling and detailed drawings in SolidWorks to hands-on prototyping and shop-floor iteration. At CubicPV, designed, built, and optimized equipment for scaling up a perovskite solar cell manufacturing line.

EDUCATION

Northeastern University — M.S. Mechanical Engineering, Certificate of Engineering Leadership Sep 2023 – May 2025
Gordon Fellow, Gordon Institute of Engineering Leadership

University of California, Berkeley — B.S. Mechanical Engineering Aug 2019 – May 2023
GPA 3.89/4.0; Cum Laude (top 10%); Dean's List, College of Engineering

University of Edinburgh — Mechanical Engineering Study Abroad Sep 2021 – May 2022

Relevant Coursework: 3D Modeling for Design, Experimentation and Measurements, Manufacturing and Tolerancing, Mechanical Behavior of Engineering Materials, Mechatronics Design, Sustainable Energy Design

EXPERIENCE

CubicPV — *Engineer III*. Bedford, MA Oct 2023 – Present
Started as Engineer II after internship, then promoted to Engineer III within a year.

Laser Scribing Automation

- Automated the laser scribing process into a manufacturable system.
- Conducted tolerance stack-up analysis and improved 6σ positional repeatability from 60 to 20 μm through targeted improvements while raising throughput 500%, yield 5%, and cutting takt time by 33%.
- Developed and built machine vision system to automatically adjust for position and skew of substrate relative to nominal.

Module Development & Efficiency

- Led module development and owned all modulization loss mechanisms, driving a 35% relative efficiency gain and a 35x degradation-lifespan improvement.
- Produced a world-record 24.0% efficiency perovskite module co-developed with NLR (formerly NREL).
- Established baseline production of scalable module processes for advanced-development manufacturing.

DMC Scanning

- Designed in SolidWorks and built a linear Data Matrix Code (DMC) scanning system using inductive sensors to automatically trigger multiple cameras, capturing each substrate's tray position in a single pass.
- Enabled tray-position tracking across three manufacturing processes, supporting positional uniformity analysis and linking substrate location during a process to its performance data.

Large-Module Photobooth

- Converted a large-area module photoluminescence box into a multi-mode, production-scale photobooth.
- Engineered and iterated a 3D-printed lighting mount, resolving a validation-stage light reflection/flare issue.

No Pollution Industrial Systems Ltd. — *Mechanical Engineering Design Intern*. Edinburgh, UK May 2022 – Jul 2022

- Designed and modeled bespoke soundproofing, heat recovery, and climate-control enclosure systems in SolidWorks for corrugated cardboard production lines, producing complex assemblies and detailed stamped sheet metal fabrication drawings.
- Built master assemblies of thousands of components nested in subassemblies with component selection across complex BOMs.
- Improved the company's pack-and-go subassembly system to streamline workflow.

MEC Team at Berkeley — *Mechanical Engineer*. Berkeley, CA Sep 2020 – Jul 2021

- Designed, prototyped, and manufactured an EVA tool attachment for NASA's Micro-g NExT Challenge.
- Editor-in-Chief of technical report employing DFMEA, FEA, and Pugh Chart to evaluate design alternatives and tradeoffs.
- Selected for Phase 2 testing at NASA's Neutral Buoyancy Laboratory.

SKILLS

Software: Arduino, AutoCAD, C++, Claude, GitHub, JMP, LabVIEW, MATLAB, Python, Simulink, SolidWorks.

Engineering: 3D Printing, CAD, Control Theory (PIDs), DAQ & Analysis, Debugging, Electromechanical Systems, GD&T, Iterative Design, Machine Shop, Machine Vision, Mechanical Design, Mechatronics, Rapid Prototyping, Tolerance Analysis

PUBLICATIONS

C60-based ionic salt electron shuttle for high-performance inverted perovskite solar modules. Science 388, 964–968 (2025). doi.org/10.1126/science.adv4701